

John Boik: “CogNarr Ecosystem: Facilitating Group Cognition at Scale”

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welcome John Bo PhD to the fourth active inference Symposium we're really pleased to have you here and looking forward to your session cogar ecosystem facilitating group cognition at scale and over to you all right thanks Ian um again my name is John Bo I'm a fellow at the Active Institute active inference Institute um and I will be talking about a cogar ecosystem to facilitate cognition at scale as Ian said uh I do want to say that uh the links for this there's a bunch of links on The Institute project page uh I think I think these slides will eventually be made public but in the meantime if you're interested you can just simply search on cogar and Kota or cogar and my last name bo and you'll find The Institute project page there's lots of links there all right so is the cogar ecosystem uh cogar means cognitive narrative and in concept it's uh it's an open- Source online extensible ecosystem uh Technologies tools libraries to facilitate group cognition uh especially for large groups and although there are many many use cases uh uh the the use case that's that one of the Prime use cases is uh deliberation strategizing Collective problem solving and decision making in the large group setting cogar essentially helps what I like to say is it helps users to tell their story that is convey their beliefs uh to a group about how the world Works relative to some situation and even more specifically convey their belief model to a group make make the belief model explicit and transparent to the group um so they might uh they might answer or address questions like what's the problem or situation what happened how did it happen What might happen next why did it happen who does it affect why who do it involve what does it mean what should be done what can be learned all of these kinds of questions um cogar allows PE allows users to tell a rich story and by Rich I mean potentially long complex nuanced Dynamic conditional uh uh um quite different quite in comparison quite different from some kind of online collaborative architecture that for example asks you to a yes or no question or ask you to give a thumbs up or thumbs down on some issue uh now this is uh this is this cogar will allow a user to convey very rich information and there's a back-end computational system that assists the digestion of

that information for the for so that the group can better understand the the many different stories so the computational system assists and guides users in construction stories uh helps the group digests the information make sense of it um uh assesses stories for their quality you know is the is the story on topic is it comprehensible that kind of thing and then the backend system also performs inference for queries that uh that various group members might have about a sto story or for a Storyteller it is important to say that the system the computational system does not predict the future that is it's not going if a bunch of people if the issue at hand say is garbage collection in a local community this the uh computational system is not going to assess stories and then predict which of them is most realistic that's not really its purpose what it what its purpose is is to convey to make transparent uh and explicit the belief of members and uh the members when they create their story when they tell their story they include what they think will happen in the future the predict they make pred the users make predictions um and also the group assesses the realism then of Any of of the stories that are submitted so what is a cogar group uh a group is is a set of people that sense their world process information anticipate comes evaluate options make decisions and in other words we can think of a group as a as a as an organism that performs cognition um and then the question is how what what design of cogar might actually facilitate functional cognition as opposed to dysfunctional cognition so examples of groups could be you know every a group informal group who like polka dot shirts uh a social club groups Civic organizations professional groups corporations nonprofits political entities such as cities and more uh so groups can range from formal to informal flexible to structured temporary to permanent small to large autocratic to democratic the the the platform is really a tool that can be used by for many use cases by many kinds of groups for many purposes there are of course some rules to keep it all functional and transparent and when I say large group what does that mean uh uh often in the literature when they talk about a lar large group uh uh deliberation and you know cognition they might be thinking of dozens of people but uh here we're thinking about perhaps dozens hundreds thousands millions there this should scale up to you know cities states and perhaps even a nation who knows you know if if I should say if cogar were ever built if it works has advertised uh you know this is all possible and and maybe I should also say at the moment this is just there's I published uh two or i' I've posted two story two articles on cogar so far two research articles and um that's all that's been done so this is this is at the beginning this is the a vision really of what this could be I'm I'm presenting a vision um so the question is what kind of design for cogar would allow some large group say a 100,000 people to to to um gather

online synchronously or asynchronously and share the potentially complex Dynamic beliefs belief models about some situation how could that be done that's the question here and I obviously I have have some ideas on how that could be done so uh but first maybe I should say that that the cogar project is inspired by active inference and in particular active inference Pro from you know from my uh from the vantage point of this project active inference provides a description of functional cognition that is as I think everyone probably knows by now watching watching the seminar that uh that cognition we can idealize it as a maybe a four-part cyclic process of predict act sense and learn and and it involves um many components for example memory recall attention um and this the entire picture of that process can be um you know that as it plays out in a human or an agent it could be either functional or dysfunctional for example uh an an agent that that cannot predict the future with accuracy is going to make decisions that are suboptimal in gen uh you know typically so that would be a a an example of dysfunctional cognition if you can't remember what happened or you can't uh predict accurately what's going to happen in the future those would all be IND indicative of dysfunctional cognition so an act of inference an agent favors choices that ensure the greatest resolution of uncertainty under the constraint as preferred outcomes are realized and all of that provides uh you know a framework such that um that the components of cognition could be recognized can each component can be um focused on as as a design element and the and the and the and the process of cognition in any group could be assessed whether whether that is a functional or dysfunctional or somewhere in between and of course that would be a you know multivalued scale so what is cogn what is cogar really um cogar essentially is designed to be part of the cognitive architecture of the group that is using it that any organism that performs cognition has some kind of cognitive architecture for for a for animal that you know involves a nerv nervous system it involves you know can for a human it can involve external objects such as Computers and Tools and um for a group it includes a set of norms rules mechanisms and institutions sensors analytic processes systems and any other procedures that facilitate or guide cognition and then Co cogar is obviously designed to be a component of the cognitive architecture of the group that uses it and viewed differently and I'll talk about this more in a moment cogar technology provides a means to digitize and make explicit a person or an agent's belief model and in that sense the the some of the technology behind cogar could be used far beyond cogar uh could be used in medicine psychology and you know unlimited number of other applications of of how you would take you know an agent or a person's belief model Make That explicit in a way that computation can occur and

um and you know include that into some kind of inference models or generative models so cogar stems from some past work that I've done and it's this past work the cogar and this past work are related in that all of it is about cognitive architecture in the past work I was looking at Social systems as cognitive architectures and by social systems I mean economic systems Financial systems governance systems I view those core societal systems as the as a means as a as a part of the cognitive archit architecture of the society that is using and that is we we think through our um governance systems we think through our economic systems we make decisions that that affect our world through our economic system for example and um the previous projects asked this question out of all conceivable societal systems designs and I don't mean conceivable in you know capitalism and socialism and the way it's been implemented in the past what I mean is like Blank Slate out of all conceivable designs for uh economic Financial governance systems which ones but might be most fit for purpose and that purpose is cognition in the active inference set so reduction of uncertainty accurate prediction of the future uh attaining those conditions that that allow for thriving of the population uh those things and I propose in this past work uh a path really of how such a project could be implemented how you know what it would do how it would go how how would you test new systems um um and so on so I referred to this earlier work and again the links are all on the cogar project page if you can find that all right so what is uh the maybe the core element of cogar is um what I call a story graph and essentially it's a meaning representation it's the it's the explicit representation of a user's belief model and this of course is a very difficult problem to solve how do you how do you take some belief model that is in a is in say the mind of of a person and put that down on paper so to speak such that computation can occur with it inference can occur with it this a very very tough problem but I think there's there's possible ways to to move forward with it um I think I mentioned or I should mention that there was two papers published on cogar first paper is more of an introduction overview the second paper is really all about it focuses on the potential meaning representations um so story graph is the sort of the core Innovation on which cogar is based and it's story graphs are used to curate understand reason with you know conduct inference on high quality information about human experiences beliefs and expectations and essentially the way it works is um a user might type in or say um some passage a few sentences perhaps a thought and uh that Passage would be translated into a what I call a story graph fragment just a small little graph of meaning graph and that little fragment would then the system would merge that fragment into the growing story graph the overall story graph and then eventually by by uh by um inputting passages

after passages after passages the story graph gets built up the meaning representation gets built up and then once it's complete or once it's almost complete then it can be translated into alternative meaning representations that might be appropriate for various you know logical models or other kinds of models probabilistic models um can maybe be translated more directly into some types of probabilistic models and could also be translated into a natural language so in in Time Event you know if kogar is successful over time it should be possible to have the system read back your story like tell me tell me you know read this back back to me and let me make sure that it's what I think what I'm intending it to be uh even you know in the far future it might be possible to make animations as an output of this you know this a little movie of this is what I think happen and this is what I think will happen in the future um so what is a meaning representation I have a picture here of a um of a abstract meaning representation that's a very common one used in the in the in the research world the the the in in natural language inference there's u in the research world there's lots of different types of of meaning representations this is just one of them I'm using as an example and this is for a story excuse me a sentence a similar a similar technique is almost always impossible to apply to other crops such as cotton soybeans and rice and you can just you I won't go into detail about this but you can kind of see that there's a graph and various words are connected uh to to each other ideas are connected to each other concepts are connected to each other and this kind of meaning representation is used throughout research and the story graph you can think of a story graph as a much more complex uh version of this it consists of layers uh you know one layer might be um predictions that the story is making about the future you know that that could be one uh layer um but uh story graph is actually more like instead of sort of static picture like this static graph a story graph is really more like a movie it's a evolving graph path really and and it involves High dimensional vectors of word meaning so there's it's far more complicated than this but this gives you a really rough idea of what I'm after uh cogner has many use cases it if if cogar were successful it would it could be ubiquitous in in in social life um so a group and there again could be all kinds of different groups many different groups Divi uh defines a group event uh its topic its purpose how how decisions were made for this event what kind of methods and rules will be used uh the group that invites people to participate and Communications in this uh for this for this group event could be many to many many to one one to many you know I have examples of all of these in the paper but um and I'll go over a couple uh shortly um use cases could be decision-making customer service emergency reports public polling instructions on how to build a you know a shelf or

something how to run a machine system descriptions of how you know how this system works educational uh U use cases and more and the later versions of cogar could include role playing strategizing um you know here's what I believe believe here's what I think this my my competitor believes and I'm going to do this you know how what would my competitor do if I did this or you know all kinds of various versions of Str how do I get the girl should I should I wear a suit and tie should I wear jeans how should I act you know like whatever the whatever the the the issue is at hand so I this is a example use case for decision-making that i' I've written out a little text here but obviously this again this would be passages that a user is inputting into the system and um and a graph is being created you know as passages are implemented so this person says you know whatever this issue X is we better first address issue y that caused X and issue y started when blank and it has the effect of blank and if we don't solve y then we can't solve a and this is likely to occur that happens and you know just on and on and on about some kind of conditional story complex conditional story and um in that in this kind of setting where there where it's a group decision making what could occur is that um uh at the top of this craft here uh story input and editing use there could be you know 10,000 users or something 10,000 members of this group they each uh create their story and and uh you know the the system would be helping them create the story too to make it easy um and then after say the first round of input the system would would perform a bunch of computations assess the what the variety of different stories are report back on the on how stories cluster with each other and all kinds of other things story quality and you know story components and all kinds of things so that the group can better understand what the picture of all these stories is and then there would be opportunity for group dialogue feedback story digestion asking questions users asking questions about different other people's uh stories and then after uh that is finished then we can go back to the beginning and uh users see everyone's story with and all the results of of Assessments and then maybe they want to change their story to to reflect what's what has been said maybe I I'm involved in this in this group and I I put down my story and then I eventually get to see what other people are saying and I go and I think oh that's a that's a good idea I want to include I want to include that piece in mind I like that or I don't like this or I'm going to comment on this one of the this part of this guy story so over a series of rounds the idea is hopefully that the belief the belief of the group would start to CES around you know core models you know but the maybe maybe one person story really captures the the group sentiment or maybe maybe the group as a whole the synthesis of their stories captures the group's sentiment and um and over a series of rounds the the thinking of

the group the belief model of the group starts to coalesce around central themes and then there could be voting at the end various kinds of voting that would be possible another use case is emergency reports again this is a text passage but it would be formed into a story graph but somebody might say hey I might call a utility and say I just I just witnessed an accident uh near this intersection a large truck uh began swerving uh it was wet shouldn't have been going so fast you know uh there have been many accidents at the corner as a kind of an aside uh truck hit an electric pole the wires started Sparks in the grass there's a school nearby uh you know all of this is there's all kinds of implications about this story right the implication is is that that there's some kind of danger maybe the maybe the the operator should send a fire truck a ambulance you know send the police you know something like that there's dangers to the there's implied dangers to the utility workers um the this is you know just a short little paragraph but it's actually quite rich and there's opportunity for many kinds of there's quite a lot of of of of information that can be inferred from this little story that isn't said explicitly um in the second paper I list a deera of about 20 uh aspects of what a what a meaning representation should be like or should be capable of doing uh I just list a few here but it should be graph-based to you know to be amenable to computation uh should be capable of handling hand Ling document length stories you know the equivalent of say you know a few pages of text or maybe even a chapter of text in a book or something like that and hand and should be able to handle a wide variety of linguistic phenomena uh should be re reducible in the sense that similar stories should make similar graphs and there's more and more and more should components should be compositional um uh and reusable and more I won't go into any more detail that but there is a whole list of of uh requirements so the last couple slides are just sort of you know fun to think about how how some of this might be done and what might happen in the future uh are there applications of category Theory here um this last paper on group interactions belief sharing via sheath and topos Theory I think is actually uh uh might be quite appropriate because that's exact what is happening in this group setting the the group is sharing their belief models those belief models are are are are sh you know the the the being being they evolve according to how the group thinking is going they're they're unifying generally the group would be unifying around one or a few stories and I think all of that could be studied mathematically uh using some ideas from category Theory but also in category Theory you know one of the ideas of category theory is to reduce the complexity of a of a project or you know system so that's important because this is obviously a complicated system uh C set property graphs might be a basis for story graphs funs might might be allow for graph

to graph graph to text and other kinds of Transformations um some ideas from category Theory might allow us to zoom in and zoom out this if you can imagine some very complicated story graph but the user just wants to focus in on just one little piece to verify that the system understood emmer her correctly and for that you might want to do a kind of a zoom in or zoom out summarization of some com some more complicated components of a story graph so there's applications for category Theory um and there's I think applications for friston's um from pixels to planning paper on renormalizing generative models um that I think there's potential there too because uh a graph really represents a path you know an evolving path of the graph itself is evolving right there's it's like a movie as I said earlier and uh so the renormalizing approach might be potentially useful there um and there's many many opportunities for agent-based agent agentic models uh for example group members understanding each other's beliefs so that's uh that's about it um yeah and again the project page has all kinds of information about this thank you John Ian if you have a question otherwise I can read one from the live chat oh yes maybe a quick one before the live chat so it's a you know lovely optimistic um sort of view that every member's story can be heard and understood and contribute to the the collective um and it's sort of making me think about you know CI citizen kind of involvement in cities and knowing what issues are important to a community um then is there a way that it can help with sort of this perennial issue of top down opinion and bottom up needs and wishes um so how would you wait how would you wait let's say the the the stories from um you know opposing members right of of a group if there's yeah right um so so there's the bigger issue that you know around what you've just said and that's how do you keep this whole thing to actually represent functional cognition like and uh obviously if one party has enormous power and another party has little power that's the you know the the cognitive process is not going to be functional so um it quite clearly a lot of work would have to go into the design of this to to prevent uh uh you know one party from dominating the entire discussion for example and there's and and work would have to go into preventing you know hijacking of the process for some you know dysfunctional means or purpose but uh it you some of that is quite addressable so it should be I I it seems to me that it would be quite possible to create a system that that is fair uh if everyone has only one voice that's one way to to prevent um one you know one person from Domina in discussion that is if this on a very simple level if there's not Bots you know if a person can't create uh imaginary Bots to be broadcasting their version of what should happen then the person would only have one voice and and over time maybe a group would get to know its members a bit and some

members would be more respected than others just naturally because they've contributed in the past they've had intelligent things to say they've you know they they are experts in this field or whatever so um we only have a minute here so I can't I can't go into much further of how that would you know the ways that kind of fairness and transparency and all that kind of stuff would be addressed but I I would think that there's many opportunities to to do so thank you John awesome presentation and looking forward to continuing the collaboration so see you e